

Discourse-JEPA: Latent World Models over Reasoning Steps

Method and five experimental cycles — fully reproducible from these slides

TextJEPA project

July 9, 2026 (repo: `TextJEPA/`, runs: `runs/disc_*`)

Idea in one slide

- ▶ Treat a **reasoning trace** as trajectories of a world model: the *state* s_t is a compressed discourse state (“what has been established”), the *action* a_t is a tiny latent code of an *intent phrase* for the next step.
- ▶ Actions state intent only (“*derive purple apples from orange jars times green pens .*”) — never the outcome (“ $= 5$ ”). The predictor $F(s_t, a_t)$ must therefore **compute consequences in latent space**.
- ▶ Generation = **search over actions**: roll candidate actions forward with F , score with an energy, execute the best in the environment, re-encode, replan (closed-loop MPC). **No decoder, no token loss anywhere** (JEPA-pure; the symbolic environment renders chosen actions via templates).
- ▶ Synthetic reasoning world $\Rightarrow$ exact ground truth for **linear probes** and **objective planning success**.

The story in five acts (red thread)

1. **A latent world model over reasoning steps works** — and we can *audit* it: predicted futures of never-taken actions match symbolic execution (Results 1, 2, 9).
2. **What silently breaks**: EMA-target prediction deletes computed content ("collusion", Result 3); the latent metric is not a progress metric (Result 5).
3. **Reconstruction-free fixes**: frozen random-init embedding targets restore content (Result 4); trajectory-geometry losses fix the metric (Results 7, 8); counterfactual *ranking* fixes the energy's discrimination (Result 10).
4. **The energy can come from the geometry itself**: a value head distilled from the latent goal distance, with zero numeric supervision, beats the fully supervised one (Results 13, 16).
5. **Baselines and limits**: likelihood selection (LMs at any tested scale/granularity) clusters far below (Result 15); failures are chain-depth-limited — the scale-up frontier (Result 16).

Glossary — every term and run-name used in this deck

discourse state s_t	one vector summarizing prompt + steps 1.. t (output of the causal state model)
predictor $F(s, a)$	MLP producing the next state from state + action code (the world model)
action a	16-d encoding of an <i>intent phrase</i> ("derive X from Y times Z") — states what to do, never the outcome
value head $V(s, s_0)$	scalar energy: estimated necessary steps remaining; the planner minimizes steps + V
value-grad	value-head gradients flow into the encoder (<code>value_detach=false</code>) so the energy shapes the representation
LDAD	Latent Displacement Action Decoding (Delta-JEPA): decode the executed action from $\Delta s = s_{t+1} - s_t$
anchor	auxiliary target: predict the <i>frozen random-init</i> encoder's embedding of the next sentence (collusion fix)
ranking (K)	margin loss ordering $V(F(s, a_i))$ over K counterfactual actions per state by their symbolic outcomes
cost ranking	same, over multi-step costs (depth + V) — calibrates search depth
hinge / monotonicity	necessary steps must decrease the latent distance to the trace-terminal state by a margin
distilled V	value head regressed on that latent goal distance (geometry target, no numeric labels)

Headline results with seed statistics (mean $\pm$ std, 3 seeds)

recipe (energy supervision)	@optimal	@slack-2
random feasible policy	0.055	0.405
base JEPA (no anchor)	0.48 ± 0.11	0.83 ± 0.07
+ frozen anchor	0.65 ± 0.03	0.89 ± 0.01
+ value-grad (= combo)	0.64 ± 0.05	0.89 ± 0.01
+ hinge (mono_hi, numeric labels)	0.63 ± 0.05	0.91 ± 0.02
+ ranking (rank_k2)	0.91 ± 0.03	0.99 ± 0.01
ranking + hinge (champion)	0.89 ± 0.02	0.99 ± 0.01
hinge + distilled V + ranking (MDR)	0.90 ± 0.03	0.98 ± 0.01
+ cost ranking (mdr_cal, look-2)	0.925 / 0.945	1.00
symbolic oracle	1.00	1.00

Single-seed numbers elsewhere in this deck sit inside these bands (± 0.03 – 0.05 @optimal, ± 0.01 @slack-2) unless flagged — *temporal straightening is the exception*: seed-fragile (0.31/0.36/0.59 @optimal).

Value-head supervision targets compared (symbolic vs geometric)

Claim: The geometric target (latent goal distance) with ranking beats numeric-label regression — IF the metric is shaped first.

V trained on	metric shaping	labels used	@opt	@slack2
numeric remaining-steps	—	scalar counts	0.64±.05	0.89
numeric + ranking	—	counts + order	0.91±.03	0.99
numeric + ranking + hinge	hinge	counts + order + binary	0.89±.02	0.99
latent goal distance	none (control)	none	0.19	0.64
latent goal distance	curvature (label-free)	none	0.30	0.72
latent goal distance	hinge	binary relevance	0.57±.05	0.89
lat. dist. + ranking	hinge	binary + order	0.90±.03	0.98

Reading: the distilled (geometric) value head is a drop-in replacement for numeric supervision once the metric it distills is goal-monotone; scalar count labels contribute nothing beyond binary relevance + ordering (0.90 vs 0.89/0.91, within seed noise).

World: iGSM-style synthetic reasoning (exact generator)

Problem sampler (rejection-sampled until $3 \leq |\mathcal{A}_q| \leq 9$; rng = Random(f"{seed}:{index}")):

- ▶ $n \sim \mathcal{U}\{6, \dots, 12\}$ variables; names = distinct (adjective, noun) pairs from 20×20 vocabulary.
- ▶ Variable i : with $p=0.35$ (always for $i < 2$) a **leaf** $x_i = c_i$, $c_i \sim \mathcal{U}\{0..22\}$; else an **internal** $x_i = x_a \circ x_b \bmod 23$, $\circ \in \{+, -, \times\}$, parents a, b sampled from $\{0..i-1\}$.
- ▶ Query q = uniform over internal variables; $\mathcal{A}_q$ = ancestors of q (incl. q) = the necessary computation; the rest are **distractors**.

Training traces: solve by repeatedly resolving a feasible variable (parents resolved); with $p=0.15$ (max 2) pick a feasible *distractor* instead of a necessary one $\Rightarrow$ off-path states for the value head. Fresh problems every epoch (train seed 1, val seed 2; 100k/5k).

A verbatim sample (val problem #3)

Prompt (definitions shuffled per-sample, question last; one sentence = one chunk):

the number of purple apples equals the number of orange jars times the number of green pens .

the number of orange jars is 15 . the number of green pens is 8 .

(+ 7 distractor definitions) how many purple apples are there ?

Solution trace — step sentence (observed outcome) vs. action intent phrase (no outcome):

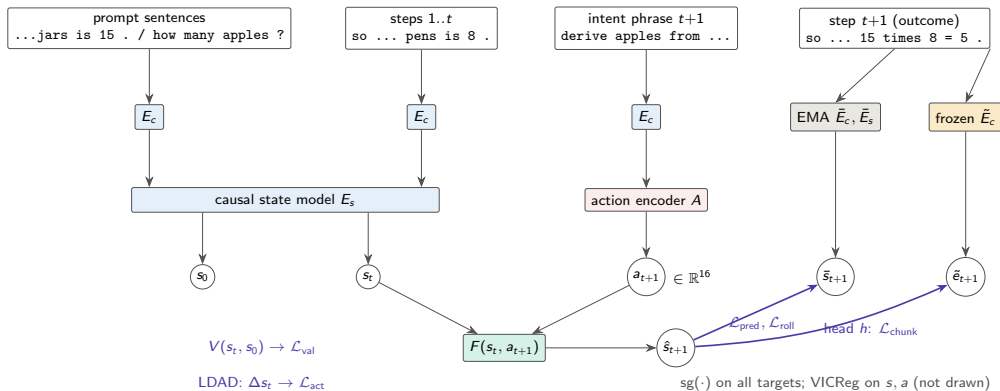
step sentence (encoder sees)	intent phrase (action encoder sees)
so the number of green pens is 8 .	look up the number of green pens .
so the number of orange jars is 15 .	look up the number of orange jars .
so the number of purple apples is	derive purple apples from orange
15 times 8 = 5 .	jars times green pens .

Note $15 \times 8 = 120 \equiv 5 \pmod{23}$: outcomes are nontrivial to predict.

Architecture (all sizes exact; 8.8–9.1M params)

- ▶ **Chunk encoder** E_c : word-level vocab (441 tokens), embed 256, 2-layer pre-norm transformer (4 heads, FF 1024), masked mean-pool $\rightarrow$ one 256-d vector per sentence. Shared for prompt, steps, intents.
- ▶ **State model** E_s : 4-layer *causal* transformer (8 heads, FF 1024) over [prompt chunks | step chunks] + learned positions + 2 segment embeddings. s_0 = hidden at the question; s_t = hidden at step t .
- ▶ **Action encoder** A : MLP $256 \rightarrow 256 \rightarrow 16$ on the intent-phrase embedding (optional FSQ, unused in main runs).
- ▶ **Predictor** F : residual MLP, $\hat{s}_{t+1} = s_t + \text{MLP}([\text{LN}(s_t); a_t])$, 2 hidden layers of 1024 (GELU).
- ▶ **Hierarchy**: CLS-transformer over $K=3$ action codes $\rightarrow$ 8-d macro action m_t ; $F_{\text{hi}}(s_t, m_t)$ predicts $\bar{s}_{t+3}$ (same latent space, HWM-style).
- ▶ **LDAD** (Delta-JEPA): MLP on $\Delta s = s_{t+1} - s_t \rightarrow$ op logits (4) + regression to the EMA intent embedding.
- ▶ **Value head** V : MLP on $[\text{LN}([s_t; s_0])]$ $\rightarrow$ scalar = predicted remaining necessary steps.
- ▶ **Teachers**: EMA copies of E_c, E_s (momentum $0.99 \rightarrow 0.999$ linear); **frozen anchor** $\tilde{E}_c$ = copy of E_c at random init, never updated.

Training pipeline



Objectives (exact losses and weights)

Let $d(x, y) = \text{smoothL1}(\text{LN}(x), \text{LN}(y))$ (per-dim mean; LN = parameter-free layer norm), masks over valid steps.

$\mathcal{L}_{\text{pred}} = d(F(s_t, a_t), \text{sg } \bar{s}_{t+1})$	weight 1.0
$\mathcal{L}_{\text{roll}} = d(F^{(t)}(s_0, a_{1:t}), \text{sg } \bar{s}_t)$ (open loop, all t)	1.0
$\mathcal{L}_{\text{hi}} = d(F_{\text{hi}}(s_t, m_t), \text{sg } \bar{s}_{t+3})$	0.5
$\mathcal{L}_{\text{vic}} = \sum_j \max(0, 1 - \text{std}(s^{(j)})) + 0.04 \ \text{offdiag}(\text{Cov}(s))\ _F^2 / D + 0.1 \text{var}(a)$	1.0
$\mathcal{L}_{\text{act}} = \text{CE}(\text{op} \mid \Delta s_t) + d(g(\Delta s_t), \text{sg } \tilde{E}_c(u_t))$ (Delta-JEPA)	2.0
$\mathcal{L}_{\text{val}} = \text{smoothL1}(V(s_t, s_0), \# \text{unresolved necessary})$	1.0
$\mathcal{L}_{\text{chunk}} = d(h(\hat{s}_{t+1}), \text{sg } \tilde{E}_c(y_{t+1})) + 0.5 d(h(\text{rollout}_t), \text{sg } \tilde{E}_c(y_t))$	0 / 2.0
$\mathcal{L}_{\text{curv}} = 1 - \cos(v_t, v_{t+1}), \quad v_t = s_{t+1} - s_t$ (straightening, arXiv:2603.12231)	0 / .02–1
$\mathcal{L}_{\text{mono}} = \mathbb{K}_{\text{nec}} [\Delta d_t^{\text{goal}} + m]_+ + \frac{1}{2} (1 - \mathbb{K}_{\text{nec}}) [-\Delta d_t^{\text{goal}}]_+$	0 / 1–2

$\mathcal{L}_{\text{chunk}}$: weight 0 in base; 2.0 in chunkpred/combo (**frozen** anchor $\tilde{E}_c$). Value head input detached except valgrad/combo. $d_t^{\text{goal}} = \text{LN-L1}$ distance of s_t to the trace-terminal EMA state; margin $m \in \{.02, .05\}$.

Training protocol

- ▶ AdamW, lr 3×10^{-4} , weight decay 0.05 (none for dim < 2 params), $\beta = (0.9, 0.95)$; cosine schedule to floor 0.05, 1000-step warmup; grad clip 5.0.
- ▶ Batch 128 problems; 20 epochs $\times$ 781 steps; **fresh data every epoch** (no memorization confound). EMA momentum $0.99 \rightarrow 0.999$ linear over training.
- ▶ Hardware: one V100-32GB, fp32, ~ 40 min/run.
- ▶ In-training eval each epoch: all losses + collapse diagnostics (per-dim std, RankMe effective rank) + closed-form ridge probes.
- ▶ Code: `scripts/train.py +experiment=...`; every run's exact config is stored in its checkpoint.

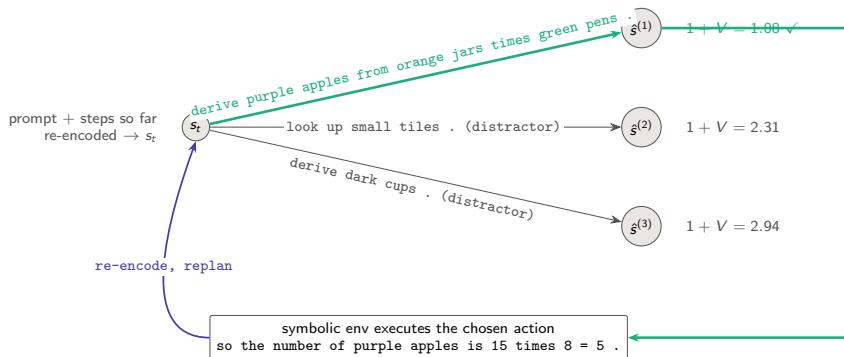
Planning: closed-loop MPC by exhaustive latent search

Interface knowledge (legitimate): feasible actions = unresolved variables whose stated dependencies are resolved — readable from the prompt. **Consequence knowledge**: latent only.

1. Encode prompt + steps so far $\rightarrow s$ (and s_0 once).
2. Enumerate feasible action *sequences* to depth L (lookahead; cap 64), encode their intent phrases $\rightarrow a_i$.
3. Roll out $\hat{s} \leftarrow F(\hat{s}, a_i)$ along each sequence.
4. Score: cost = steps taken + $V(\hat{s}_{\text{end}}, s_0)$ (estimated total steps).
5. Execute the first action of the argmin in the symbolic environment $\rightarrow$ real outcome sentence appended; go to 1.

Success = query resolved within the **strict optimal budget** $|\mathcal{A}_q|$ (slack 0: any distractor pick is fatal). 200 val episodes. Discrete enumeration replaces CEM; a sampled action prior would need CEM/MPPI.

Planning: search graph (one replanning round, lookahead 1)



V values are predicted *remaining necessary steps* of the predicted next latent; lookahead 2 scores 2-step latent rollouts.

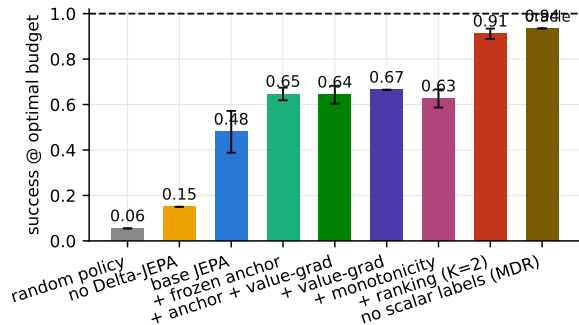
Experiment grid (each = one training run, 20 epochs)

run	$\mathcal{L}_{\text{chunk}}$ anchor	value grad $\rightarrow$ encoder	other
base	—	no	reference
no_delta	—	no	$\mathcal{L}_{\text{act}}$ weight 0
chunkpred	frozen, w=2.0	no	—
valgrad	—	yes	—
combo	frozen, w=2.0	yes	—
value_only	—	yes	<i>all</i> JEPA weights 0
frozen_enc	—	no	encoders frozen at init
straight- $\{\text{lo}, \text{mid}, \text{---}\}$	frozen, w=2.0	yes	+ $\mathcal{L}_{\text{curv}}$, $\lambda \in \{.02, .05, .1\}$
mono / _hi	frozen, w=2.0	yes	+ $\mathcal{L}_{\text{mono}}$, w1 m.02 / w2 m.05
geo_lo / _geo	frozen, w=2.0	yes	both regularizers
mono_novalue	frozen, w=2.0	no	+ $\mathcal{L}_{\text{mono}}$, <i>no value head</i>
rank- $\{\text{k2}, \text{k4}\}(\text{_nodelta})$	frozen, w=2.0	yes	+ ranking over K counterfactual actions
champion	frozen, w=2.0	yes	ranking K=2 + $\mathcal{L}_{\text{mono}}$ w2
$\{\text{shuffled}, \text{no_distract}, \text{size*}\}$	frozen, w=2.0	yes	grounding controls (Results 9, 11)

Baselines at plan time: **random feasible policy**, **symbolic oracle** (=100%), and **oracle goal-distance energy** (LeWM-style, uses the encoded solved state — diagnostic only).

Result 1 — training objectives determine plannability

Claim: The anchor lifts the base model decisively; value-grad and the hinge sit within seed noise of it; counterfactual ranking is the step change. Bars: mean $\pm$ std over 3 seeds (thin caps).



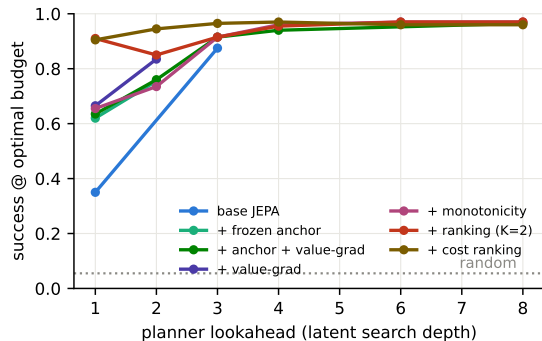
Success within the *strict optimal budget*, greedy lookahead 1, 200 episodes, mean necessary steps 4.07.

With slack 2 (mean of 3 seeds): base 0.83, +anchor 0.89, +hinge 0.91, +ranking / MDR **0.98–0.99** (random 0.41).

Distractor pick rate: random 0.43 $\rightarrow$ base 0.28 $\rightarrow$ valgrad 0.14.

Result 2 — search depth is a planning-time compute knob

Claim: Success grows with search depth up to a shared ≈ 0.97 ceiling: rollouts of F and the energy stay trustworthy over many imagined steps, and depth even rescues weak energies.



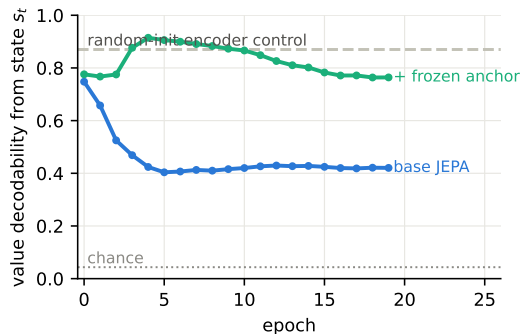
Lookahead d = feasible d -step sequences (capped), scored at the imagined end state; no retraining. Depth curves are single-seed; the depth-1 anchors carry ± 0.03 – 0.05 seed bands (leaderboard slide).

All recipes converge to ≈ 0.96 – 0.97 by depth 4 — the residual is the deep-chain failure mass (Result on failures), not search.

Base JEPa: $0.35 \rightarrow 0.87$ (depth 3); `mono_hi` $0.655 \rightarrow 0.97$ (depth 6). The `rank_k2` dip at depth 2 is the calibration anomaly; cost ranking (brown) removes it and dominates at every depth.

Result 3 — encoder–predictor collusion, measured

Claim: Pure state-prediction JEPA *removes* outcome information that was linearly present at initialization; a frozen outcome anchor prevents it.



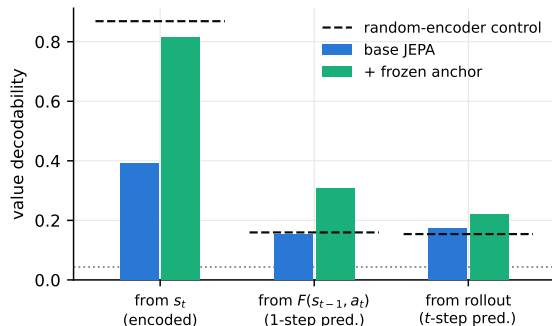
Probe: linear decodability of the just-computed value (23 classes) from s_t , val split.

Both loss sides are learned; EMA targets *trail the online encoder*, so deleting a hard-to-predict feature lowers the loss — stop-grad does not pin content.

Consequence in base: answer decodable from the terminal state at only 0.26 (random-init control: 0.65).

Result 4 — latent arithmetic under the frozen anchor

Claim: With $\mathcal{L}_{\text{chunk}}$, the predictor computes outcomes it never observes: value decodability from *predicted* latents doubles the random-encoder control.



Dashed = random-init encoder control (surface memory, no computation); dotted = chance $1/23$.

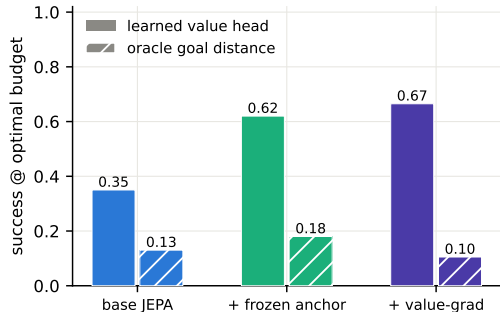
From $F(s_{t-1}, a_t)$: base 0.15 (= control) $\rightarrow$ anchor **0.31**.

Answer from terminal state: 0.26 $\rightarrow$ **0.98**.

Full-trace open-loop rollout: 0.22 (error compounds; matches Result 2's need for re-encoding).

Result 5 — the goal energy must be learned (LeWM test)

Claim: Raw latent distance to the oracle goal state plans barely above random; the trained value head is the energy that works.



Oracle goal = encoded terminal state of the *true* solution (planner gets it for free — upper-bound diagnostic).

Scoring $\|\text{LN}(\hat{s}) - \text{LN}(s_{\text{goal}})\|_1$: base 0.13, anchor 0.18, valgrad 0.105 (distractor rates $\approx$ random).

$\Rightarrow$ *unregularized* JEPA geometry is not a distance-to-goal metric. Result 7 shows temporal straightening fixes exactly this.

Result 6 — Delta-JEPA ablation; stability

Claim: Displacement-level action decoding is load-bearing for planning (replicates Delta-JEPA, arXiv:2606.31232, in language).

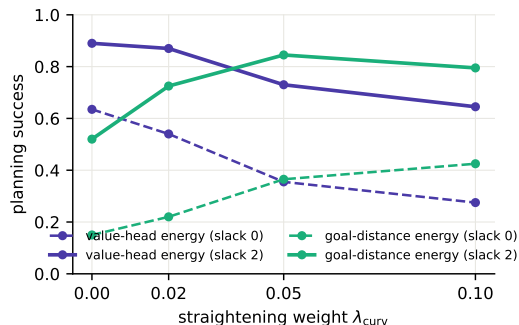
	success@opt	success@slack2	distractor rate	op from Δs
base	0.35	0.75	0.28	1.00
no_delta	0.15	0.51	0.40	0.98*

*op stays decodable (it correlates with surface features) but transitions lose the *action-discriminative* structure the planner needs.

Stability (all runs): per-dim state std ≈ 1.03 , RankMe effective rank 220–243 / 256, no collapse; VICReg + EMA sufficed, discrete codes (FSQ) not needed.

Result 7 — temporal straightening makes raw geometry plannable

Claim: The curvature loss (arXiv:2603.12231) turns latent distance into a progress metric: goal-distance planning goes $0.15 \rightarrow 0.43$ @optimal ($0.52 \rightarrow 0.85$ @slack 2) — but it *trades off* against the value head.



Dose-response over $\lambda_{\text{curv}} \in \{0, .02, .05, .1\}$ (all on the combo recipe):

	0	.02	.05	.1
velocity cos	-.25	-.01	.42	.67
corr(d^{goal} , remaining)	.73	.75	.80	.86
value probe from s_t	.89	.89	.74	.43

Geometry improves exactly as content decodability degrades; $\lambda = .05$ is the raw-geometry sweet spot.

Result 8 — two energies, one metric: the content–geometry trade-off

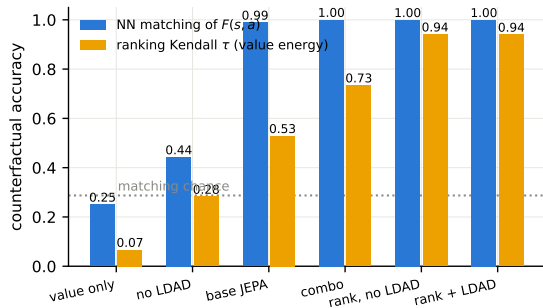
Claim: Straightening buys geometry planning, the monotonicity hinge keeps content and sets the value-planning record; geometry needs JEPA losses, not value labels.

run	goal-distance energy		value-head energy	
	@opt	@slack2	@opt	@slack2
combo ($\lambda = 0$)	0.15	0.52	0.635	0.89
straight_mid ($\lambda=.05$)	0.365	0.845	0.355	0.73
mono_hi (hinge w2 m.05)	0.255	0.68	0.655	0.935
mono_novalue (<i>no V</i>)	0.20	0.65	—	—
value_only (<i>no JEPA</i>)	0.06	—	0.075	0.43
random	0.06	0.095	0.06	0.095

Three readings. (i) One latent metric cannot currently maximize both energies — a learned *projection head* for the geometric energy is the obvious fix (tested in Result 10: it softens but does not remove the trade-off). (ii) A model with **no value head** still plans at 0.65 @slack 2 from pure geometry. (iii) value_only's geometry is *exactly random*: planning-relevant structure comes from the JEPA objectives, not the value labels.

Result 9 — the predictor is counterfactually grounded (direct audit)

Claim: From each state we enumerate *all* feasible actions, predict their next-state latents, and check against symbolic execution the model never saw: $F(s, a)$ is correct per action, and LDAD / outcome anchors are what ground it.



Audit metrics (100 episodes, val):

nearest-neighbor *matching* of $F(s, a_i)$ to the executed-and-encoded true next states; Kendall τ of the value energy vs. true post-action remaining steps.

Training data shows **one action per state** — grounding comes from cross-problem generalization alone.

Shuffled-action control: matching $0.30 \approx$ chance. RSA of pairwise next-state geometry: 0.93 (base+LDAD) vs 0.48 (no LDAD).

Result 10 — counterfactual ranking: the strongest lever

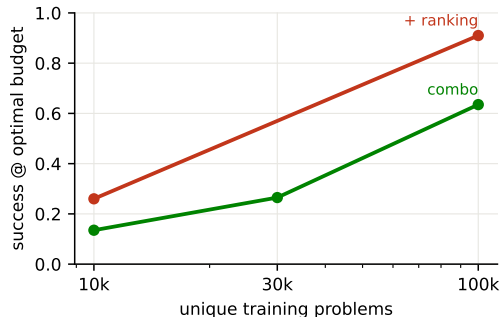
Claim: Hinge-ranking K alternative actions per state by symbolic outcome nearly closes the gap to the oracle; $K=2$ suffices; it can replace LDAD but not data diversity; it trades search-depth calibration for 1-step precision.

	@opt	@slack2	@opt look-2	value τ
combo	0.635	0.89	0.76	0.73
rank_k2	0.910	0.985	0.85 ↓	0.94
rank_k4	0.925	0.98	0.85 ↓	0.94
rank_k4 <i>no LDAD</i>	0.875	0.965	—	0.94
combo <i>no LDAD</i>	0.48	0.805	—	0.69
champion (rank+mono)	0.905	0.99	—	0.96

↓ lookahead-2 *hurts* ranking models (0.91→0.85) while helping regression values (mono_hi 0.655→0.735): the margin loss perfects 1-step ordering but distorts V 's absolute scale that multi-step cost sums need. **champion** also has the best raw geometry (τ_{goal} 0.82, oracle-goal planning 0.655, no curvature loss). Seed replicates: rank_k2 0.885–0.91 @opt, 0.98–0.985 @slack2.

Result 11 — data diversity is the counterfactual fuel

Claim: Grounding comes from cross-problem coverage: shrinking unique problems collapses planning; ranking supervision sharpens but cannot substitute.



Same optimizer steps, fewer unique problems:

problems	10k	30k	100k
combo @opt	0.135	0.265	0.635
+ ranking	0.26	—	0.91
matching	0.82	1.00	1.00
value τ	0.22	0.44	0.73

Matching survives small data; *ranking quality* does not — the predictor stays roughly right, the energy loses discrimination. Within-trace variation matters too: no distractors $\Rightarrow$ 0.635 $\rightarrow$ 0.415.

Result 12 — depth calibration: CostRanking and macro jumps

Claim: Plain ranking breaks lookahead (0.91→0.85); two independent fixes exist, and they don't stack because they repair the same defect.

	look-1	look-2	look-3 flat	look-3 F_{hi}
rank_k2	0.910	0.850 ↓	0.915	0.970
rank_cal (+CostRanking)	0.905	0.945	—	0.915
champion_cal	0.910	0.945	—	0.915

CostRanking: rank full MPC costs across depths — executed 2-step continuation ($2 + V(F^2)$) vs 1-step alternatives ($1 + V(F(a'))$); zero new data. **Macro jumps:** score 3-step sequences with one F_{hi} call — worse *fidelity* than F^3 (matching 0.62–0.70 vs 0.77–0.91) but stays on V 's training distribution. 1000-episode: rank_cal look-2 = 0.946 vs hier 0.931 — calibration wins once you have it; macro jumps are the no-retraining workaround. Train-side hierarchy loss: neutral (combo_nohier 0.670/0.910).

Result 13 — the non-symbolic value head (headline)

Claim: Shape the metric with the hinge, distill it into V , add ranking: best planner with no numeric value labels anywhere.

run	energy supervision	@opt	@slack2
champion_cal	<i>numeric</i> + ranking	$0.91 \pm .01$	0.995
mono_distill_rank	binary only (geometric V)	$0.90 \pm .03$	$0.98 \pm .01$
mono_hi	<i>numeric</i> regression	$0.63 \pm .05$	$0.91 \pm .02$
mono_distill	binary (no ranking)	$0.57 \pm .05$	0.89
mono_lf_distill	<i>none</i> (all-steps hinge)	0.355	0.775
geometric	<i>none</i> (curvature)	0.300	0.715
distill_v	<i>none, unshaped</i> (control)	0.190	0.640

Rows 1 vs 2 are the direct **numeric-vs-geometric** comparison at matched search settings: the geometric value head (V regressed on the latent goal distance d^{goal} , ordered by binary-outcome ranking) matches the numeric-label champion within seed noise ($0.90 \pm .03$ vs $0.91 \pm .01$). Shaping is required: unshaped distillation fails (0.19). Mean $\pm$ std over 3 seeds where shown; single-seed otherwise.

Result 14 — stability: the Delta-JEPA claim does not transfer; VICReg is load-bearing

Claim: LDAD alone does not prevent collapse in language JEPA, and VICReg is not redundant even with EMA + stopgrad + frozen anchor.

run	EMA	sg	VICReg	result
ldad_pure	—	—	—	collapse (std .027, rank 80)
ldad_pure_sg	—	✓	—	collapse (0.11 @opt)
combo_novic	✓	✓	—	0.27 @opt (vs 0.635)
combo	✓	✓	✓	0.635 @opt

Honest negative replication of arXiv:2606.31232’s stability claim in our domain (their evidence is visual control; our traces/action space differ). Also: non-residual predictor $\hat{s} = \text{MLP}([\text{LN}(s); a])$ performs at least as well as residual (0.710/0.920) — the residual skip is convenience, not mechanism; LDAD constrains *encoder* displacements, not predictor outputs, so it is not trivialized by either choice.

Result 15 — the baseline ladder (action-selection parity)

Claim: Every likelihood/reconstruction selector — token- or sentence-level, any tested scale — clusters at 0.47–0.74; explicit counterfactual training adds 20–30 points on top of all of them.

model (selection energy)	params	@opt	@slack2
sentence-LM (decoder CE)	7.2M	0.470	0.810
sentence-LM+latent tgt (decoder CE)	7.2M	0.510	0.800
sentence-LM+latent tgt (latent dist.)	7.2M	0.605	—
token LM (likelihood, best lr)	8.2M	0.670	0.910
naive JEPA (combo, value head)	9.1M	0.635	0.890
token LM	25.5M	0.735	0.940
JEPA + ranking (MDR / +cal)	9.1M	0.935	0.99–1.00

Parity protocol: every model scores the *same* feasible candidates' rendered step sentences (LM: log-likelihood; sentence-LM: decoder CE or latent distance; JEPA: latent energy), same budget. LR swept for both families (LM prefers 10^{-3} , JEPA 3×10^{-4}). Note the within-model dissociation: the sentence-LM's *latent* energy outranks its own decoder (0.605 vs 0.510) — selection is a latent-geometry problem even for reconstruction-trained models.

Result 16 — the final recipe and its ablation matrix

Claim: $\text{disc_mdr_cal} = \text{anchor} + \text{LDAD} + \text{VICReg} + \text{hinge} + \text{distilled } V + \text{ranking} + \text{cost}$
ranking: **0.925 / 1.000 @slack-2 / 0.945 @look-2**. With ranking present the recipe is redundant — every single removal costs only 2–7 points.

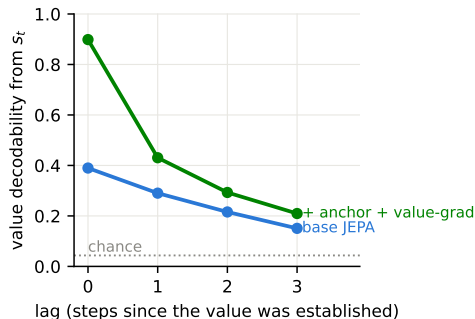
variant	@opt	@s2	τ_V	m	variant	@opt	@s2	τ_V	m
+cost rank	.925	1.00	.96	1.0	–VICReg	.895	.99	.90	.99
MDR (s1/2/3)	.87–.935	.97–.99	.95	1.0	–hierarchy	.910	.98	.95	1.0
–anchor	.865	.99	.93	.99	–hinge	.915	1.00	.94	1.0
–LDAD	.865	.97	.92	.94	–distill	.895	.99	.91	.98

Interactions worth knowing: VICReg is catastrophic without ranking (0.27) but nearly free with it — ranking gradients hold the space open. A *non-residual* predictor makes LDAD planning-redundant on combo (0.71 = 0.71) and is $\geq$ residual everywhere. Replacing state-prediction targets with anchored observation prediction fails outright (obs_only: matching 0.22) — both prediction levels are required.

Failure analysis (500 eps): champion failures are chain-depth-limited — 4.4% (3–4 steps), 16% (5–6), 59% (7–9); distractor-count effects are depth confounds. Depth is what the scaled domain stresses next.

Probes I — what the discourse state actually is

Claim: A recency-weighted working memory over a relational scaffold: structure is objective-independent, content is objective-dependent.

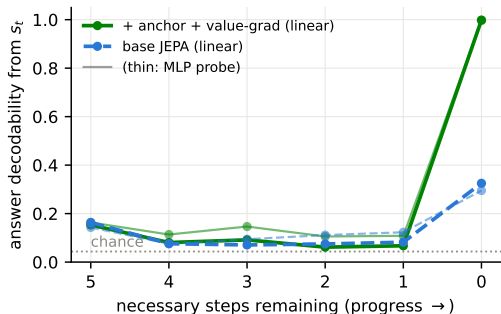


probe (combo / all recipes)	acc	maj.
resolved-set member $[s_t; \text{oh}(v)]$	0.82	0.67
ancestor-of-query $[s_0; \text{oh}(v)]$	0.76	0.56
var just resolved (from Δs)	0.33	0.19
query identity (from s_0)	0.19	0.16

These rows are **identical across every training recipe** (base, no-LDAD, geometry, ranking) while value decodability swings 0.39–0.90: the encoder learns the scaffold from text alone; the objectives decide what content survives. Relevance is encoded *relationally* (0.76), not as a query pointer (0.19).

Probes II — emergence, memory, circular code

Claim: No anticipatory answer computation: the answer appears exactly when its variable resolves — and only collusion-free models can hold it.



Emergence is a step function: 0.06–0.18 mid-trace (thin MLP lines: small nonlinear surplus), 1.00 the moment the query resolves; base manages only 0.33 even then (collusion).

Memory decays with lag (Probes I): $0.90 \rightarrow 0.43 \rightarrow 0.29 \rightarrow 0.21$ (chance 0.04) — a working buffer, not a symbol table.

Circular coding: ridge R^2 for $\cos(2\pi v/23)$ from s_t : 0.52 (combo) vs 0.20 (base) — the mod- p value is partly geometric, and the circle sharpens under the anchor fix.

Probe suite (champion vs. reference; random-encoder control)

probe (linear, val split)	base	chunkpred	random-enc
value of current step from s_t	0.39	0.81	0.87
value from $F(s_{t-1}, a_t)$ (latent arithmetic)	0.15	0.31	0.15
value from open-loop rollout	0.17	0.22	0.15
op type from Δs	1.00	1.00	0.97
step goal-relevance from Δs	0.89	0.93	0.89
remaining steps from s_t	0.44	0.61	0.41
resolved count from s_t	0.97	0.92	0.95
final answer from terminal state	0.26	0.98	0.62
answer from prompt alone s_0 (should be hard)	0.035	0.05	0.04
value-head MAE (remaining steps)	0.76	0.57	—

combo adds value-grad: MAE **0.38** with the same collusion-free probes (state value 0.89).

Scale-up preview — the hard domain and faithful iGSM

Claim: On 10–18-step problems everything collapses — the likelihood selector hardest — and the ranking/capacity/search recipe degrades most gracefully. This is the quantified frontier for

the next phase.	hard domain (14–24 vars, 10–18 steps)	@opt	@slack2	look-2	look-4
	reference recipe (combo, 9M)	0.000	0.070	0.035	0.175
	minimal-labels + cost rank (mdr_cal, 9M)	0.120	0.445	0.105	0.150
	same recipe, 35M	0.090	0.410	0.195	0.315
	token LM (8M, parity selection)	0.010	—	—	—

Capacity only pays *through search* (35M: 0.09 greedy → 0.32 at look-4). Separately, the pipeline now runs on **100%-faithful iGSM** (the official facebookresearch generator vendored; our env drives the reference renderer; full solutions pass the official checker 60/60 med, 99/100 hard) — first faithful-domain training runs completed, planning evaluation in flight.

Honest limitations & what this is not

- ▶ **iGSM-style, not iGSM**: flat DAG, binary mod-23 ops, one question template, templated English — no hierarchical categories / instance-vs-abstract parameters of the original benchmark.
- ▶ **Actions are observed intents**, not variational latents: the symbolic world offers an enumerable interface. The variational prior $p(a \mid s, g)$ (for CEM/MPPI over sampled actions) is future work.
- ▶ **The best goal energy is still supervised** (remaining-steps labels); straightened raw geometry closes much of the gap (0.845 vs 0.935 @slack 2, Results 7–8) and the curvature loss is *label-free*, but the monotonicity hinge uses the same labels as the value head.
- ▶ Value labels shape only the value head unless `valgrad`; the op-CE in LDAD uses the executed action's type (JEPA-legit: actions are observed at training time).
- ▶ Text “generation” is action selection + template rendering; a learned renderer is orthogonal to the world model.

Next steps

- ▶ **Depth-calibrated ranking:** rank multi-step rollout costs, not only 1-step candidates, so lookahead helps ranking models too (Result 10's anomaly); target ≥ 0.95 @optimal with look-2.
- ▶ **Error analysis of the last 10%:** what do the champion failures share (chain depth, op mix, tie patterns)?
- ▶ **Hierarchical planning:** use the trained F_{hi} + macro codes to propose 3-step jumps, refine with F (HWM).
- ▶ **Unobserved actions:** variational $q(a \mid s_{t-1}, s_t)$ with KL to prior $p(a \mid s, g)$; inspect emergent action clusters.
- ▶ **Harder worlds:** faithful iGSM (hierarchy + category sums), larger moduli, paraphrase templates, natural-language noise; scale.